

Question Paper

Exam Date & Time: 27-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Data Structures and Algorithms [ECE_4447]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Differentiate between Program and Algorithm. Discuss the specifications of an algorithm. (4)
A) Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Differentiate between Space complexity and Time complexity. (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Represent and discuss the significance of Sparse matrix transpose. (3)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Illustrate the conversion of Infix to Postfix operation by taking any arithmetic expression. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Differentiate between Stacks and Queues. (3)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- C) With an example code, explain the insertion operation at the end for a Singly linked list. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) Enlist the terminologies used in Binary tree with an example. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) With example, explain Inorder Traversal method used in Binary tree. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Expand the terminologies used in AVL tree with proper definitions. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) With an illustration, explain the adjacency list to represent graphs. (4)
A) Course Outcome: "CO4" / Cognitive Level: "Application"

B) Suggest an algorithm for Linear search considering both Best case and Worst case complexities. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

C) Discuss the steps of Merge sort algorithm. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

5) Suggest an algorithm for Merge sort highlighting the significance of the same. (4)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) Discuss the steps involved in heap sort. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

C) Discuss the necessity of Breadth First Search algorithm. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECE_4447:Data Structures and Algorithms

#	Course Outcome	Description	Marks
1	1	CO 1	4
2	2	CO 2	6
3	3	CO 3	20
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECE_4447:Data Structures and Algorithms

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	10(20%)
3	Application	8	27(54%)
4	Analysis	4	13(26%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)